

Fall 2020

Applied quantitative Methods for Non-quantitative doctoral researchers

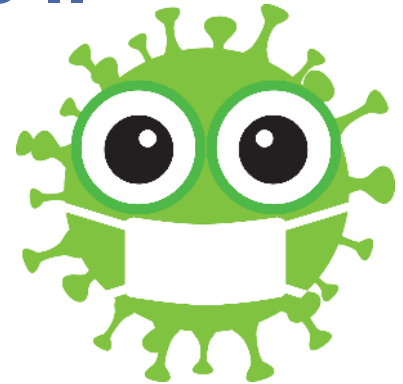
Session 09: Regression analysis II

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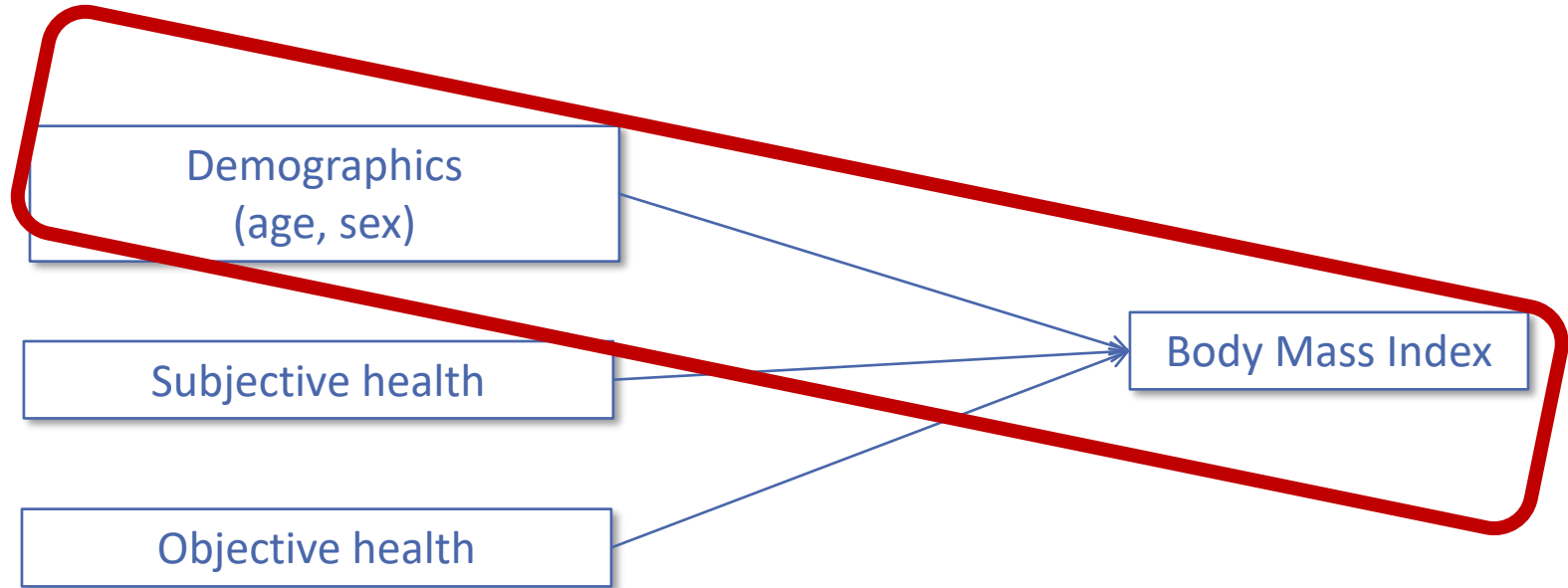
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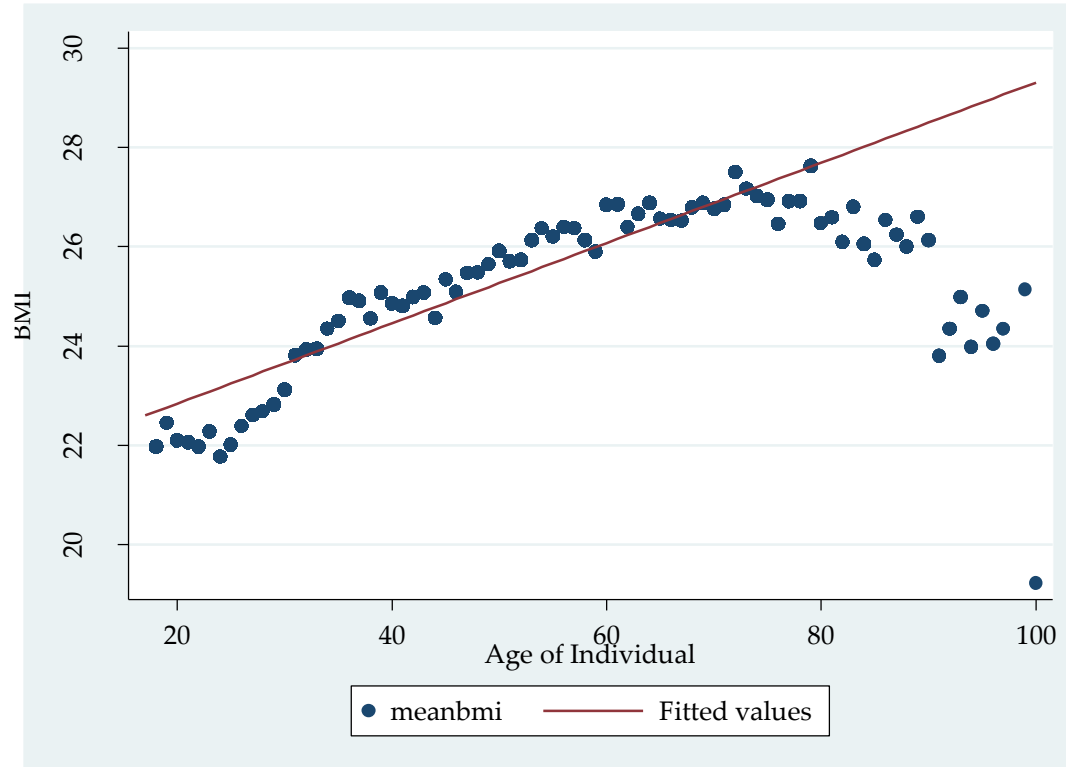
Our example again



Regression analysis

h_bmi	Coef.	Std. Err.
age	.0809353	.0017636
_cons	21.21955	.0937372

Interpretation: with every year we estimate BMI to increase by 0.08



Modelling complex relationships

Bivariate OLS assumes a linear relationship that is the same for everyone. However, often:

- Effects are not linear
- Vary by subpopulation
- Dependent on other characteristics.

Examples:

- Gender wage gap
- Economic theory of diminishing marginal returns
- Income effects on happiness
- etc.

Non-linear relationships

```
. reg h_bmi c.age
```

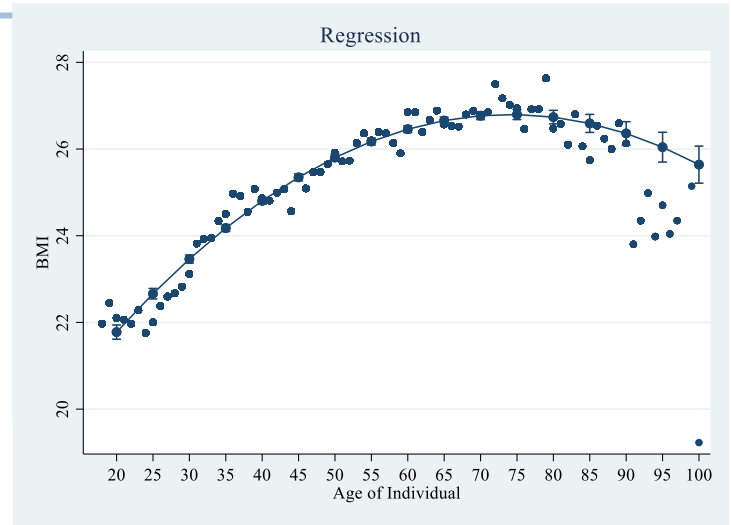
Source	SS	df	MS	Number of obs	=	18,122
Model	37107.2831	1	37107.2831	F(1, 18120)	=	2106.07
Residual	319260.396	18,120	17.6192272	Prob > F	=	0.0000
				R-squared	=	0.1041
				Adj R-squared	=	0.1041
Total	356367.679	18,121	19.6660052	Root MSE	=	4.1975

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0809353	.0017636	45.89	0.000	.0774784	.0843921
_cons	21.21955	.0937372	226.37	0.000	21.03582	21.40329

```
. reg h_bmi c.age#c.age
```

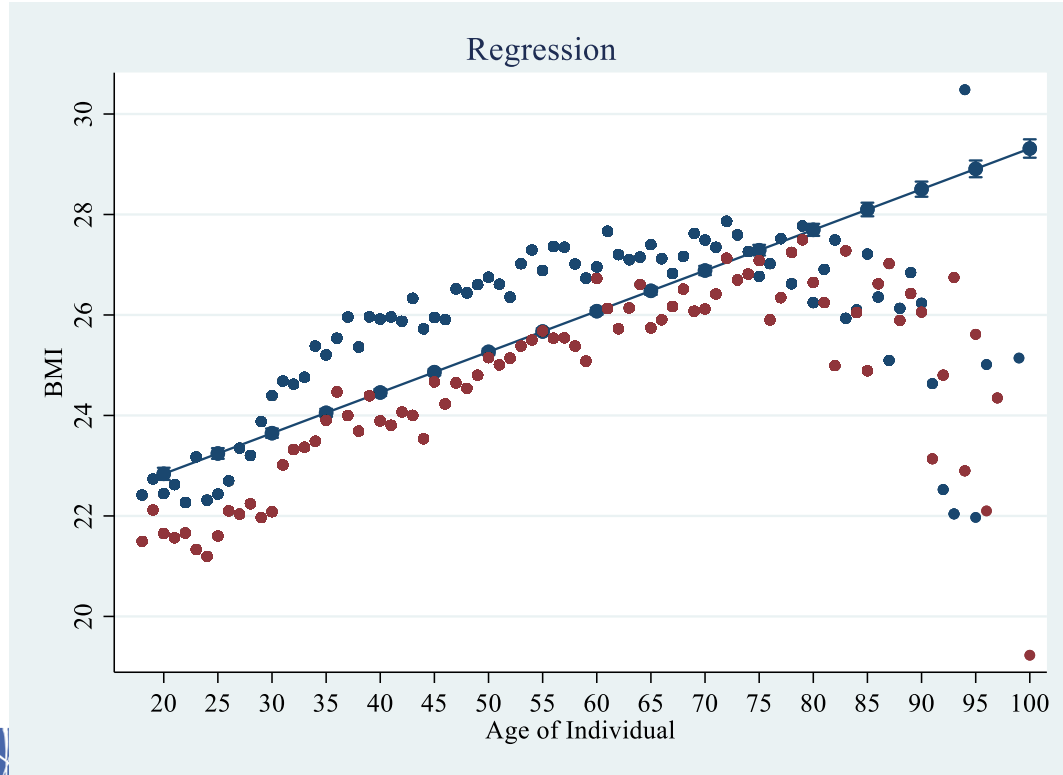
Source	SS	df	MS	Number of obs	=	18,122
Model	43093.531	2	21546.7655	F(2, 18119)	=	1246.21
Residual	313274.148	18,119	17.2898145	Prob > F	=	0.0000
				R-squared	=	0.1209
				Adj R-squared	=	0.1208
Total	356367.679	18,121	19.6660052	Root MSE	=	4.1581

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.2545518	.0094927	26.82	0.000	.2359451	.2731584
c.age#c.age	-.0017188	.0000924	-18.61	0.000	-.0018999	-.0015378
_cons	17.37288	.2266268	76.66	0.000	16.92867	17.81709



$$bmi_i = \beta_0 + \beta_1 * age_i + e_i$$

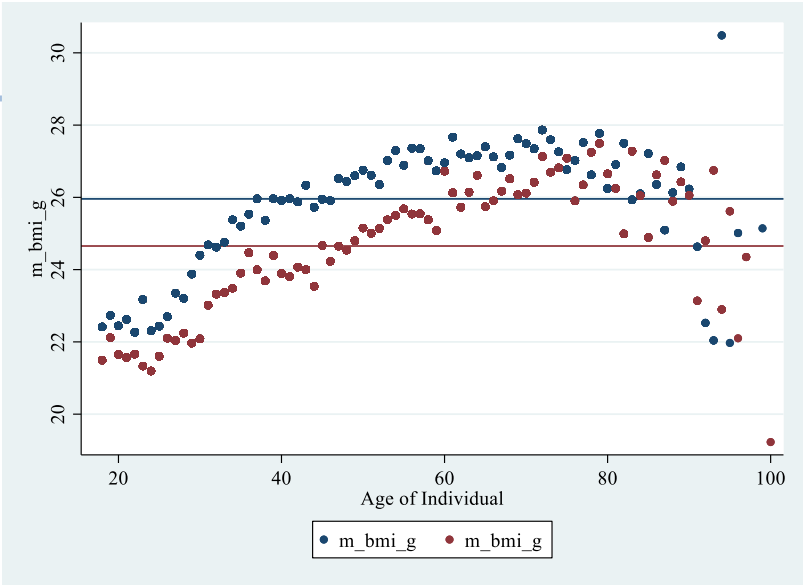
$$bmi_i = \beta_0 + \beta_1 * age_i + \beta_2 * age_i^2 + e_i$$



```
. reg h_bmi man
```

Source	SS	df	MS	Number of obs = 18,122		
Model	7683.90201	1	7683.90201	F(1, 18120) = 399.31		
Residual	348683.777	18,120	19.2430341	Prob > F = 0.0000		
				R-squared = 0.0216		
				Adj R-squared = 0.0215		
Total	356367.679	18,121	19.6660052	Root MSE = 4.3867		

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
man	1.303623	.0652376	19.98	0.000	1.175751	1.431495
_cons	24.65368	.0450872	546.80	0.000	24.5653	24.74205



$$bmi_i = \beta_0 + \beta_1 * man_i + e_i$$

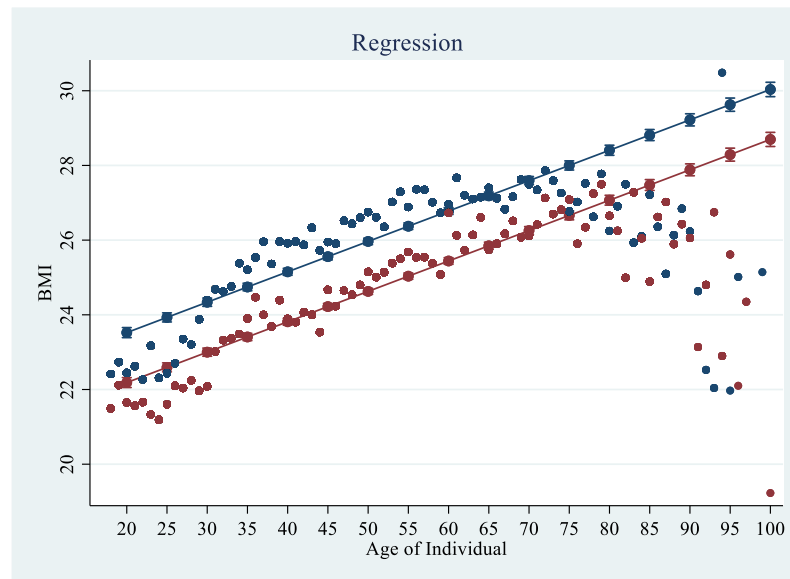
with man: 1 for males and 0 for females

Modelling more complex relationship

```
. reg h_bmi age man
```

Source	SS	df	MS	Number of obs = 18,122		
Model	45197.567	2	22598.7835	F(2, 18119) = 1315.90		
Residual	311170.112	18,119	17.1736913	Prob > F = 0.0000		
Total	356367.679	18,121	19.6660052	R-squared = 0.1268		
				Adj R-squared = 0.1267		
				Root MSE = 4.1441		

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0813829	.0017413	46.74	0.000	.0779699	.084796
man	1.337746	.0616344	21.70	0.000	1.216936	1.458555
_cons	20.55814	.0974326	211.00	0.000	20.36716	20.74911



$$bmi_i = \beta_0 + \beta_1 * age_i + \beta_2 * man_i + e_i$$

Interaction/moderation effects

```
. reg h_bmi c.age##i.man
```

Source	SS	df	MS	Number of obs = 18,121
Model	45285.7011	3	15095.2337	F(3, 18118) = 17.17
Residual	311081.978	18,118	17.1697747	Prob > F = 0.0000
Total	356367.679	18,121	19.6660052	R-squared = 0.1272
				Adjusted R-squared = 0.1268

h_bmi	Coef.	Std. Err.	t	P> t
age	.0851265	.0024003	35.46	0.000
man				
[1]male	1.733572	.1852598	9.36	0.000
man#c.age				
[1]male	-.0079001	.0034869	-2.27	0.023
_cons	20.36975	.1280825	159.04	0.000



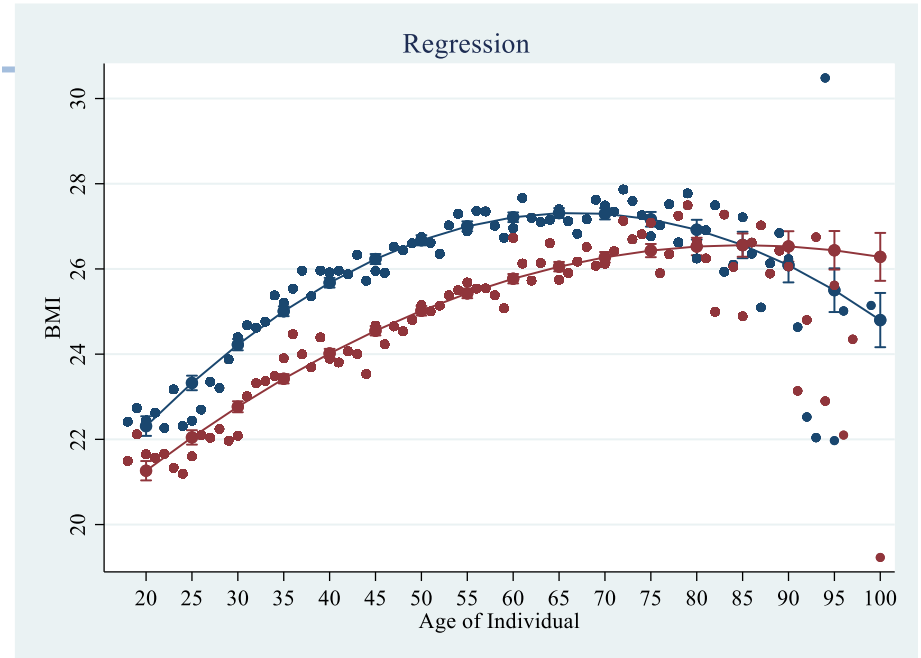
$$bmi_i = \beta_0 + \beta_1 * age_i + \beta_2 * man_i + \beta_3 * (age_i * man_i) + e_i$$

Careful regarding the interpretation: β_2 alone now indicates the difference between man and women at the age of 0. Not meaningful nor actually in the data

```
. reg h_bmi i.man#c.age#c.age
```

Source	SS	df	MS	Number of obs	=	18,122
Model	51827.3444	5	10365.4689	F(5, 18116)	=	616.60
Residual	304540.335	18,116	16.8105727	Prob > F	=	0.0000
Total	356367.679	18,121	19.6660052	R-squared	=	0.1454
				Adj R-squared	=	0.1452
				Root MSE	=	4.1001

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
man						
[1]male	-.3946357	.4478716	-0.88	0.378	-1.272507	.4832352
age	.2125533	.0128778	16.51	0.000	.1873117	.237795
man#c.age						
[1]male	.0928781	.01879	4.94	0.000	.0560479	.1297084
c.age#c.age	-.0012484	.000124	-10.07	0.000	-.0014914	-.0010053
man#c.age#c.age						
[1]male	-.0010377	.0001831	-5.67	0.000	-.0013966	-.0006787
_cons	17.51167	.3108877	56.33	0.000	16.9023	18.12104



$$bmi_i = \beta_0 + \beta_1 * age_i + \beta_2 * man_i + \beta_3 * (age_i * man_i) + \beta_4 * (age_i * age_i) + \beta_5 * (age_i * age_i * man_i) + e_i$$

```
. reg h_bmi satlife
```

Causality

Source	SS	df	MS	Number of obs	=	18,066
				F(1, 18064)	=	144.91
Model	2827.69592	1	2827.69592	Prob > F	=	0.0000
Residual	352492.497	18,064	19.5135351	R-squared	=	0.0080
				Adj R-squared	=	0.0079
Total	355320.193	18,065	19.6689838	Root MSE	=	4.4174

h_bmi	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
satlife	-.2238148	.0185926	-12.04	0.000	-.2602581	-.1873715
_cons	26.82489	.1324953	202.46	0.000	26.56519	27.08459

```
. reg satlife h_bmi
```

Source	SS	df	MS	Number of obs	=	18,066
				F(1, 18064)	=	144.91
Model	449.229024	1	449.229024	Prob > F	=	0.0000
Residual	55999.6071	18,064	3.10006682	R-squared	=	0.0080
				Adj R-squared	=	0.0079
Total	56448.8361	18,065	3.12476259	Root MSE	=	1.7607

satlife	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
h_bmi	-.0355569	.0029538	-12.04	0.000	-.0413466	-.0297672
_cons	7.802391	.0758108	102.92	0.000	7.653795	7.950988

```
. reg satlife h_bmi
```

Source	SS	df	MS	Number of obs	=	18,066
				F(1, 18064)	=	144.91
Model	449.229024	1	449.229024	Prob > F	=	0.0000
Residual	55999.6071	18,064	3.10006682	R-squared	=	0.0080
				Adj R-squared	=	0.0079
Total	56448.8361	18,065	3.12476259	Root MSE	=	1.7607

satlife	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
h_bmi	-.0355569	.0029538	-12.04	0.000	-.0413466	-.0297672
_cons	7.802391	.0758108	102.92	0.000	7.653795	7.950988

```
. reg satlife h_bmi h_hstat
```

Source	SS	df	MS	Number of obs	=	18,030
				F(2, 18027)	=	2208.25
Model	11085.4337	2	5542.71687	Prob > F	=	0.0000
Residual	45247.8732	18,027	2.51000572	R-squared	=	0.1968
				Adj R-squared	=	0.1967
Total	56333.3069	18,029	3.12459409	Root MSE	=	1.5843

satlife	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
h_bmi	.0088269	.0027461	3.21	0.001	.0034444	.0142095
h_hstat	.8361859	.0128456	65.09	0.000	.8110073	.8613646
_cons	3.896143	.0909002	42.86	0.000	3.71797	4.074317

Controls

```
. pwcorr satlife h_bmi h_hs
```

	satlife	h_bmi	h_hstat
satlife	1.0000		
h_bmi	-0.0892	1.0000	
h_hstat	0.4341	-0.2483	1.0000

Controlling for confounding variables

Most research is interested in causal attribution: X causes Y

Using regression can suffer from two important short-comings:

- Reverse causality: Y (party) causes X
 - Example: Well-being & BMI
- Confounding: X correlates with Y partly/only because a third variable Z correlates with X and Y independently
 - Example: „Ice cream sales (X) causes people to drown in their pools (Y)“ -> Z?

Confounding variables

Controlling for all confounding variables would solve the problem

However „all confounding variables“ are rarely:

- Known
- Observed

Example: BMI & well-being could be linked through many pathways: genes, social & physical environment, personality traits, etc.

- Specifying the „correct“ controls depends on the research question and data availability

Example of good and bad controls

RQ: *Does life satisfaction decline with age?*

satlife	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	-.0049523	.0007002	-7.07	0.000	-.0063247	-.0035799
_cons	7.215924	.0370356	194.84	0.000	7.143331	7.288517

satlife	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
age	.0149441	.0006818	21.92	0.000	.0136077	.0162804
h_hstat	.9147922	.0125821	72.71	0.000	.8901302	.9394541
_cons	3.157405	.0648325	48.70	0.000	3.030328	3.284482

Selection as a confounding issue

Particularly researcher in Economics are interested in estimating the causal effect of certain programs or events. E.g. the causal effect of your PhD on earnings, unemployment programs, etc.

RQ: *Do hospitals make you healthier?*

Possible regression approach:

$$\text{current_health}_i = \beta_0 + \beta_1 * \text{hospital_visit}_i + e_i$$

hospital_visit_i measured as at least one hospital visit in the last 6 months

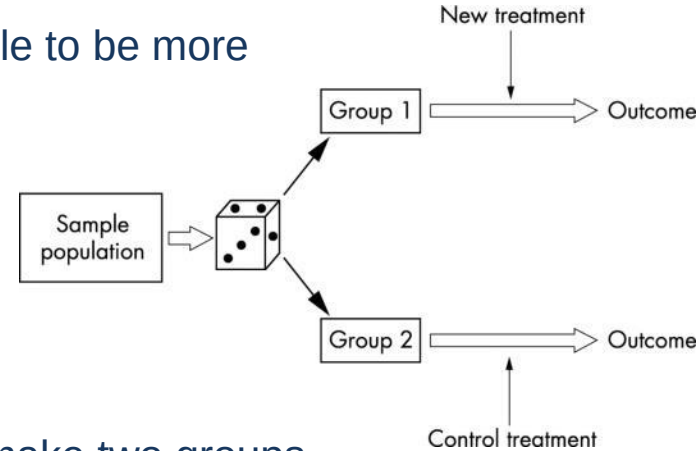
Experimentation for causal inference

In our case, we want to know if more income causes people to be more satisfied:

$$bmi = \beta_0 + \beta_1 employment + e$$

We could conduct an experiment:

- Select a sample of unemployed people and randomly make two groups
- Give people in one “the treatment” group a job, but not the “control” group.
- If the sample is large enough bmi (reverse causality) and all other (omitted) variables should be equal.



Randomization to balance all possible confounders

Imagine a random sample of the Danish unemployed population ($n=2,000$). Now 1,000 are randomly assigned to each group either treatment (employed) or control (unemployed).

- Both groups are a random sample of the population of interest
- We can expect the mean of confounding variable Z (e.g. age) to reflect the population mean for each group.
- All Z s are balanced between the groups and therefore not explain a difference in BMI even though they might be correlated.
- All Z s are independent of employment status as this was randomly assigned.

$$bmi = \beta_0 + \beta_1 employment + e$$

- No control variables are required as we excluded ALL confounding variables by design

Caution: this hypothetical experiment would be practically unfeasible (forcing people not to get jobs) and might not fully reflect employment relationships in the real-world which heavily rely on selection into different kind of employment

Application

Let's try this...

The end...



Furthermore, I consider that we must do more to reduce human impact on the environment.



... might be near!